



Faculty of engineering, Cairo University
Electronics and Electrical Communication Department
ELC4046 – Process Control



Lab 1 _ PROCESS DYNAMIC CHARACTERISTICS

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Submitted to :

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LABORATORY EXERCISE 1

PROCESS DYNAMIC CHARACTERISTICS

OBJECTIVE: To become familiar with various forms of process dynamic characteristics, and to learn a method of constructing a simple process model from step test data. **Optional:** To become familiar with obtaining data from frequency response tests.

PREREQUISITE: Completion of Laboratory Exercise 0 Getting Familiar or an equivalent amount of familiarity with the program operation.

BACKGROUND: All self-regulating processes have both steady state and dynamic characteristics. From a process control standpoint, the most important characteristic is the process gain. That is, how much does the process variable (PV) change for a change in controller output. If both the PV and the controller output are expressed as normalized variables (i.e., 0 - 100%), then the process gain is a dimensionless number.

The two most important dynamic characteristics of a process are the amount of dead time in the process and its time constant. Real processes rarely exhibit a response of a pure first order lag (time constant) and dead time, but can often be approximated as a first order lag and dead time.

This exercise tests for the process gain, dead time and time constant for both a “pure” process (can be exactly represented as first order lag plus dead time) and for a more realistic process which can only be approximated as a first order lag plus dead time.

1. RUNNING THE PROGRAM

Run **PC-ControlLAB**.

Slow down the simulation speed via **View | Scroll Rate** and select a mid-value.

2. FIRST ORDER LAG PLUS DEAD TIME PROCESS

Click on **Process | Select Model**.

Highlight "Folpdt.mdl" (First Order Lag Plus Dead Time) and press **Open**.

Press **Zoom** and change the PV scale range to 50-75. (Note that the PV scale has already been converted to 0 – 100% of measurement span.)

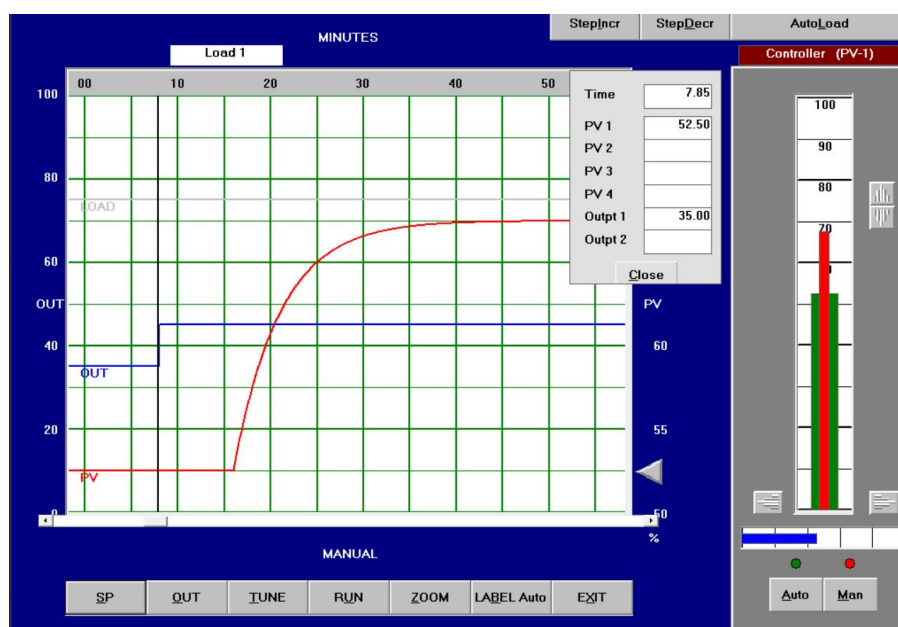
With the controller in MAN, press **Out**. Note the initial values:

Present PV:

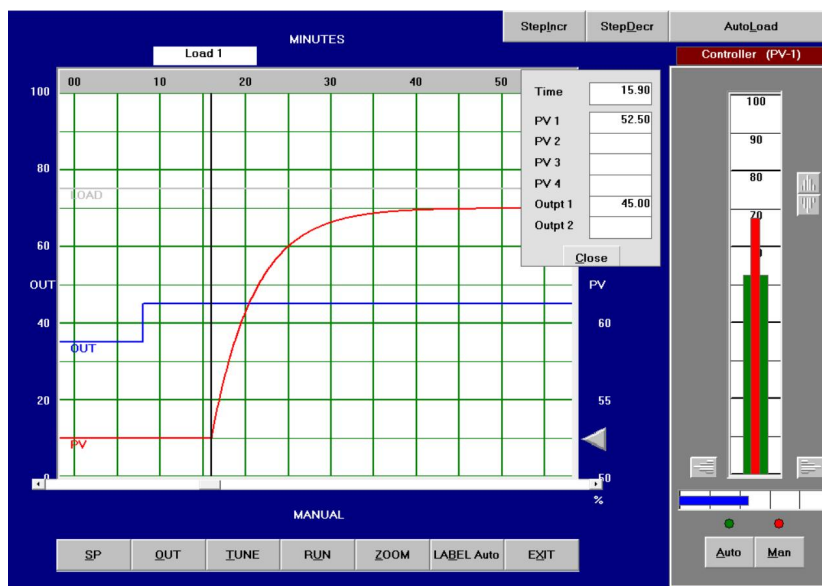
52.5

Present Controller Output:

35.00



Key in a new output value of 45.00



(Hint: After the numerical value is keyed in, wait until a vertical green line on the grid is just crossing the grid boundary before pressing ENTER. This will make it easier to estimate subsequent times.)

After the PV has stabilized at a new value, press **PAUSE**.

Final value of PV:

67.49

What type of process response does this appear to be?

Self regulating (first order lag plus deadtime FOPDT)

How much did the PV change?

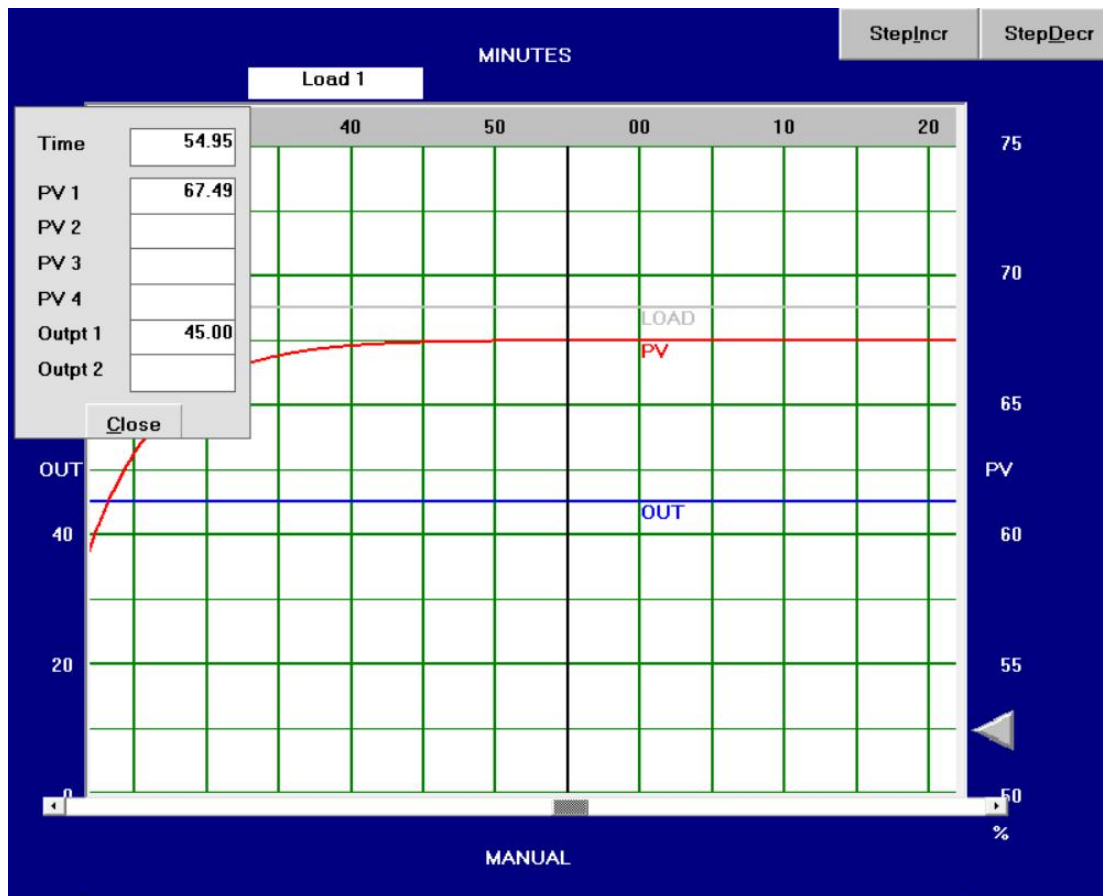
67.49-52.50=14.99%

How much did you change the controller output:

45-35=10%

Process Gain: $K_P =$

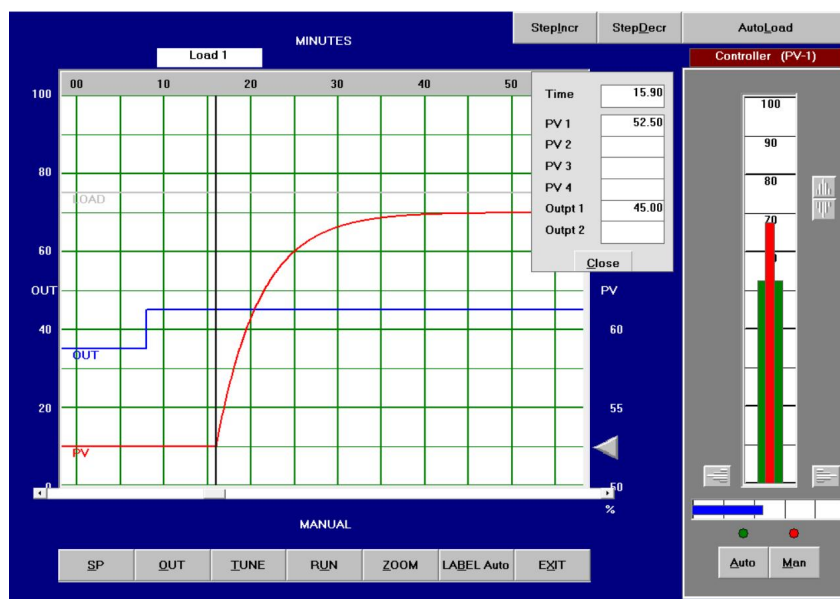
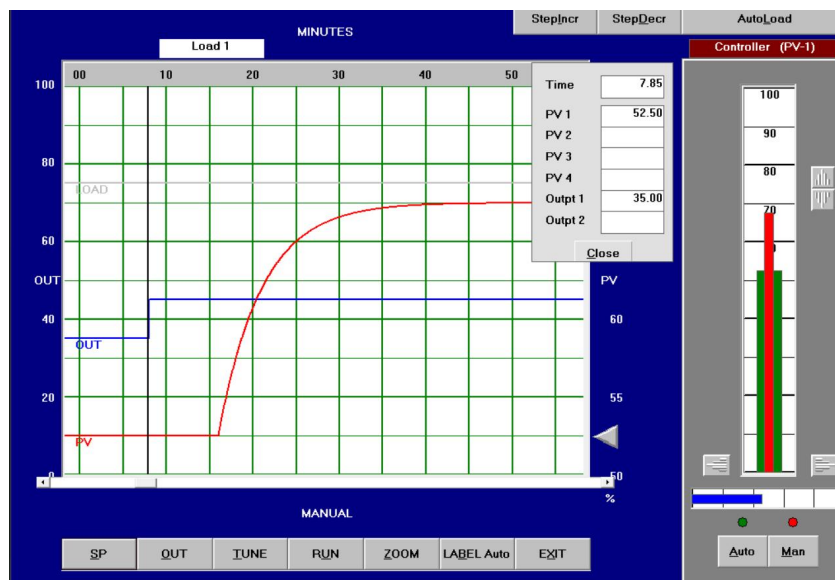
14.99% / 10%=1.499



How long after the controller output change before the PV started changing?

Dead Time (θ)

_____ (15.90min-7.85min)=8min _____

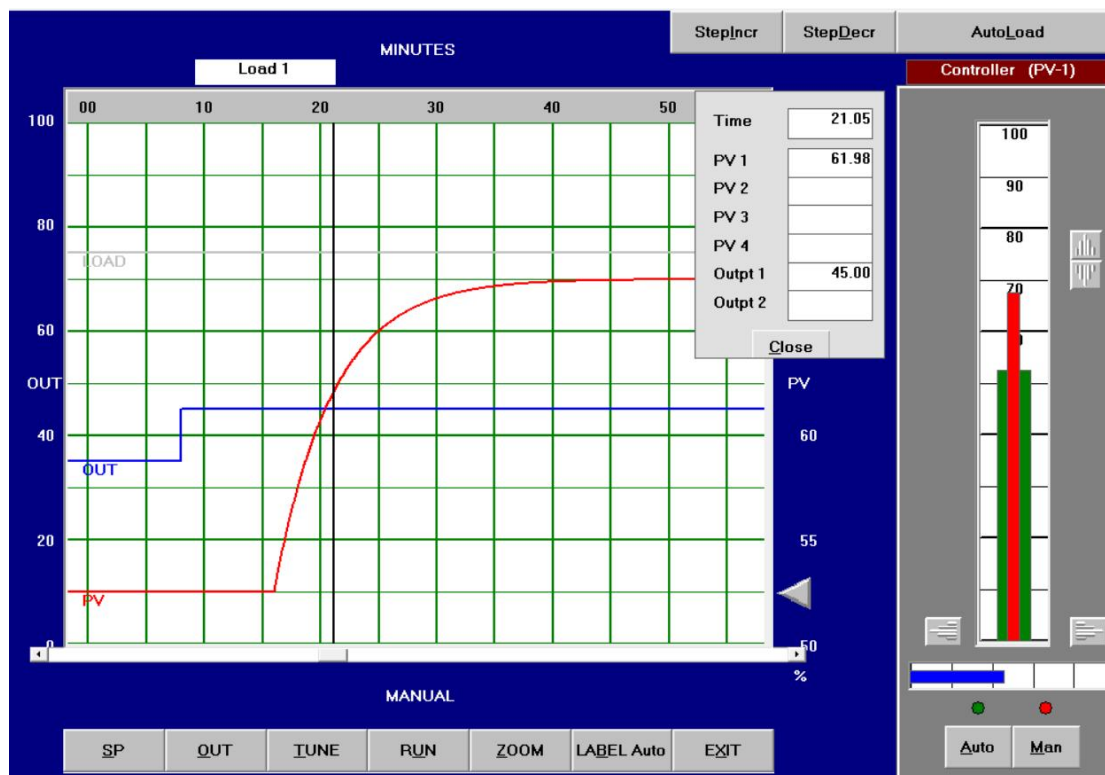


Calculate 63.2% of PV change: $(0.632 \cdot 14.99) = 9.47368$

Actual value of PV at 63.2% of change: $(0.632 \cdot 14.99) + 52.5 = 61.97$

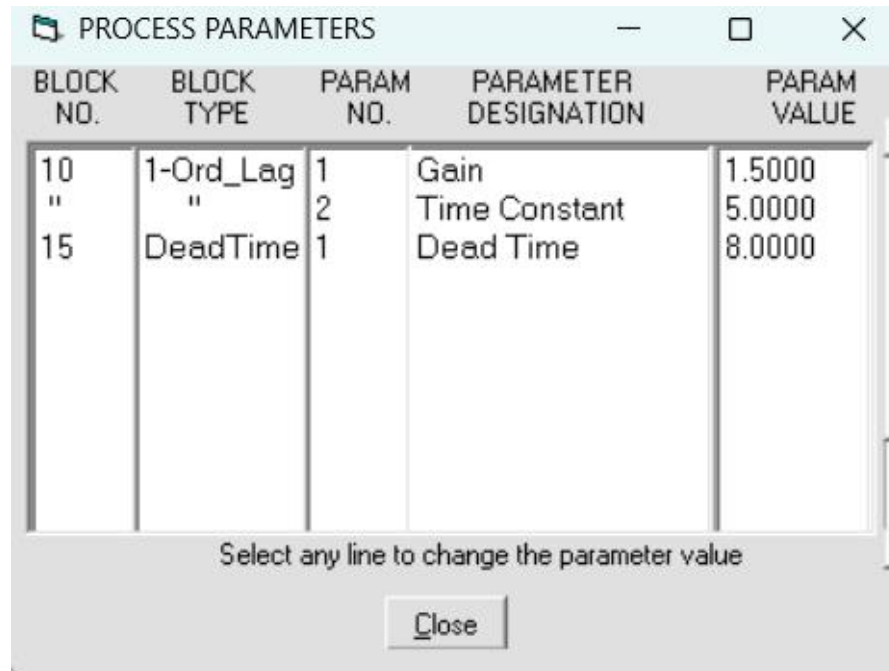
How long after the PV started changing (i.e., at the end of the dead time) before the PV crossed the 63.2% point?

Time Constant (τ) $21.05 - 15.90 = 5.1$



Select **Process | Change Parameters**. Observe the values listed for Dead Time and Time Constant. Do these parameter values agree with what you observed? ____

yes, the parameters in the list is similar to the calculated parameters. _____



BLOCK NO.	BLOCK TYPE	PARAM NO.	PARAMETER DESIGNATION	PARAM VALUE
10	1-Ord_Lag	1	Gain	1.5000
"	"	2	Time Constant	5.0000
15	DeadTime	1	Dead Time	8.0000

Select any line to change the parameter value

Close

Select **Process | Initialize**.

Select **Process | Change Parameters**.

Select "Dead Time" and change its value to 2.0 (minutes).

Select "Process Gain" and change its value to 1.0.

Select "Time Constant" and change its value to 3.0 (minutes).

Press **CLOSE**.

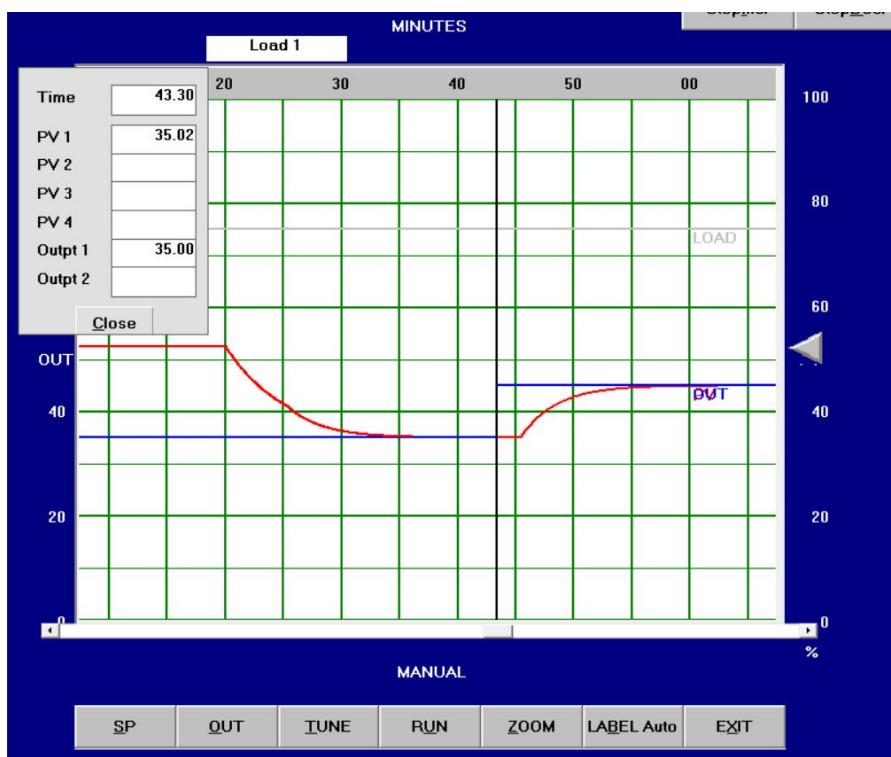
PROCESS PARAMETERS				
BLOCK NO.	BLOCK TYPE	PARAM NO.	PARAMETER DESIGNATION	PARAM VALUE
10	1-Ord_Lag	1	Gain	1.0000
"	"	2	Time Constant	3.0000
15	DeadTime	1	Dead Time	2.0000

Select any line to change the parameter value

Close

Change the controller output from 35.0 to 45.0. Observe the response. Is this what you would expect?

The PV is equal to the MV, which is expected because the process gain is 1. The system also responds faster because the dead time and the time constant are both lower than in the previous case



3. UNKNOWN PROCESS

Click on **Process | Select Model**. Highlight “Generic” and press **Open**.
 Notice that the PV scale is now in Engineering Units, rather than in percent.
 (If not, then select **View | Display Range | Engineering Units**)

Upper end of scale (corresponds to high end of transmitter range) 500

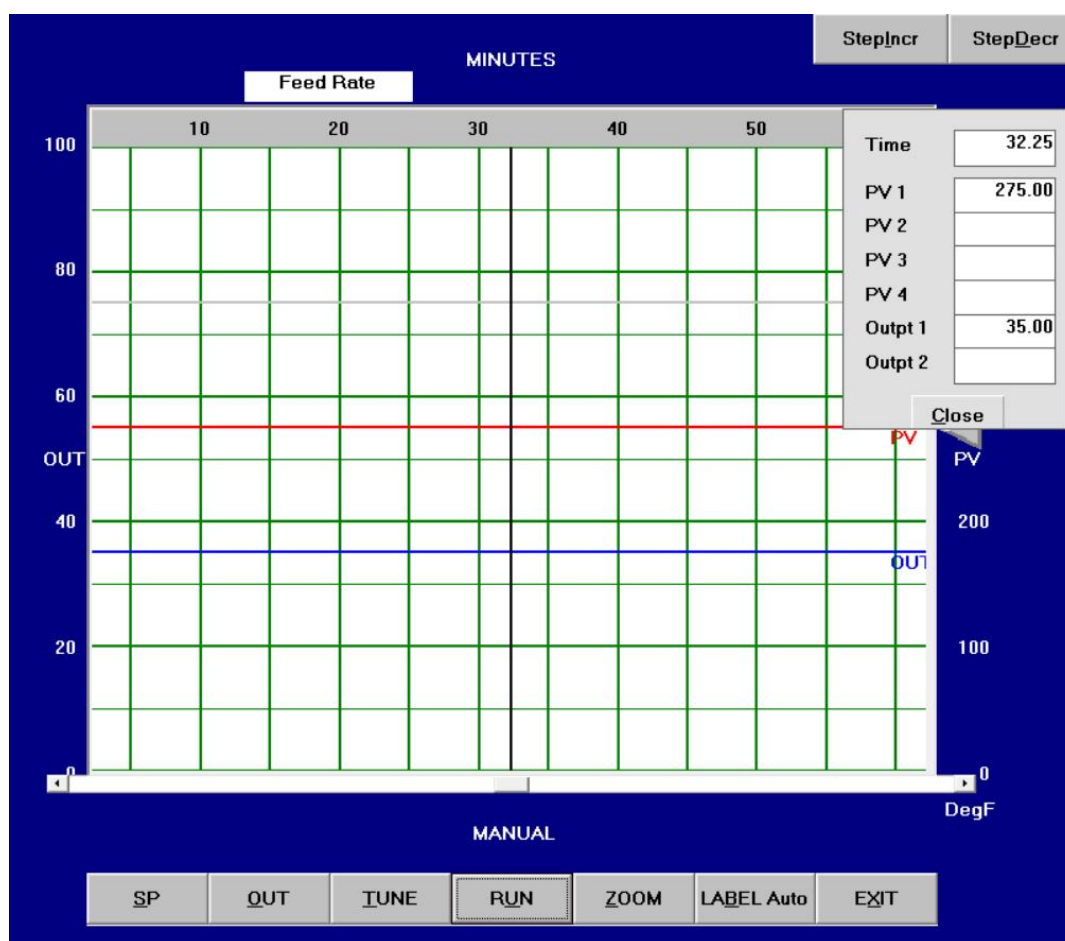
Low end of scale (corresponds to low end of transmitter range) 0

Span of PV 0:500

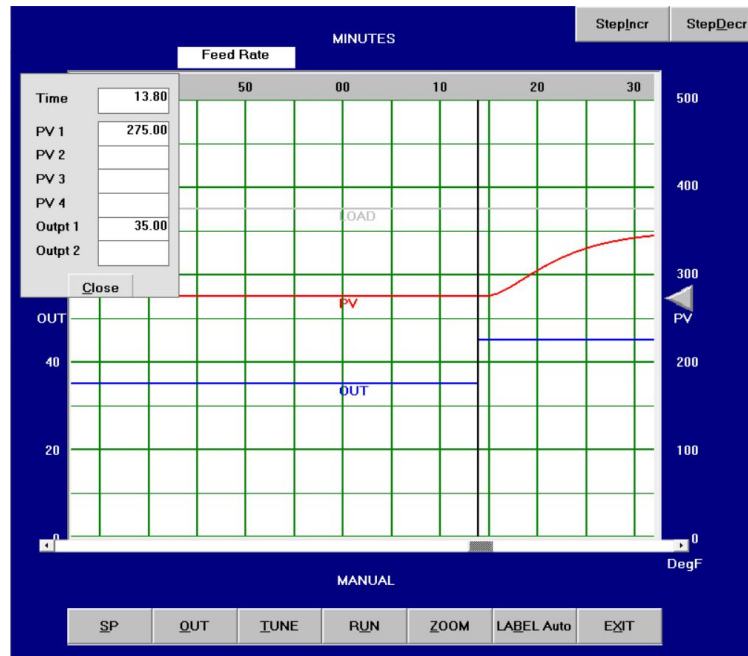
Initial value of PV (in engineering units) 275.0

Initial value of PV (in percent of span) $(275/500)*100=55\%$

Controller Output: 35.0



Change the controller output from 35.0 to 45.0.



When the PV reaches (apparent) equilibrium, press **Pause**. Does this look like a true first order lag plus dead time process? No

Does it look “approximately” like a first order lag plus dead time process?

yes , It is approximately like FOPDT but it is not a true FOPDT because the steepest tangent is not at the starting point.

What is the final value of the PV (to the nearest whole number)?

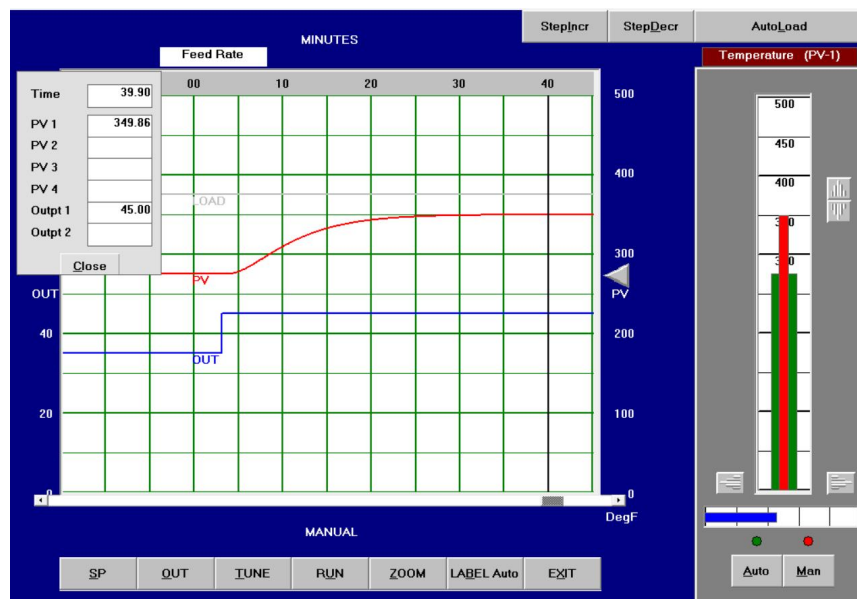
349.86

How much did the PV change, in engineering units?

349.86-275=74.86

How much did the PV change, in percent of span?

(74.86/500)*100=14.97%



Estimate the process gain.

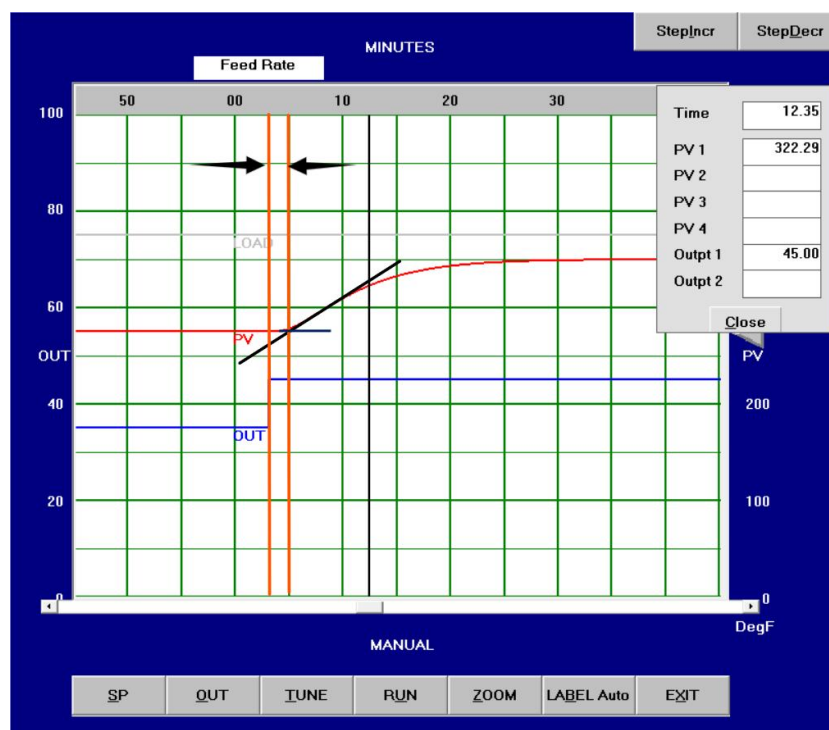
 K_P (0.1497)/(0.1)=1.497

To estimate the dead time, draw (or visualize) a tangent to the PV curve, drawn at the point of steepest rise. From the time of controller output change to the intersection of this tangent with the initial steady state value is the apparent dead time (see Figure 1).

Apparent dead time:

 θ 1.5 min

Different observers might estimate anywhere between 1½ to 2 minutes. For the purpose of calculating controller tuning parameters, it is better to take the longer value where there is any uncertainty, since that will produce more conservative controller tuning.



The apparent time constant is the time from the end of the dead time to 63.2% of the process rise

Apparent time constant:

 τ 7.85 min

Calculate the value of the PV at 63.2% of the complete PV change, then use the scroll bar to determine the time precisely between the (estimated) end of the dead time and the 63.2% point at the PV response.

NOTE: One of the uses that can be made of the estimates of process gain, dead time and time constant is to calculate controller tuning parameters. (See Laboratory Exercise 9, PID Tuning from Open Loop Tests.) Since dead time is more difficult to control than a lag, then if you estimate dead time too short, you are estimating that the process is easier to control than it really

is. This will result in controller tuning parameters that cause the loop to be overly aggressive. Similarly, if you estimate the time constant too long, you are estimating that the process is easier to control than it really is, and again the resulting controller tuning parameters will cause the loop to be overly aggressive. On the premise that if one is to make an error, it is better to err in the conservative direction than in the aggressive direction, then the following pragmatic guidelines can be given:

If there is any uncertainty in estimating the process parameters,

*estimate the dead time on the long side, and
estimate the time constant on the short side.*

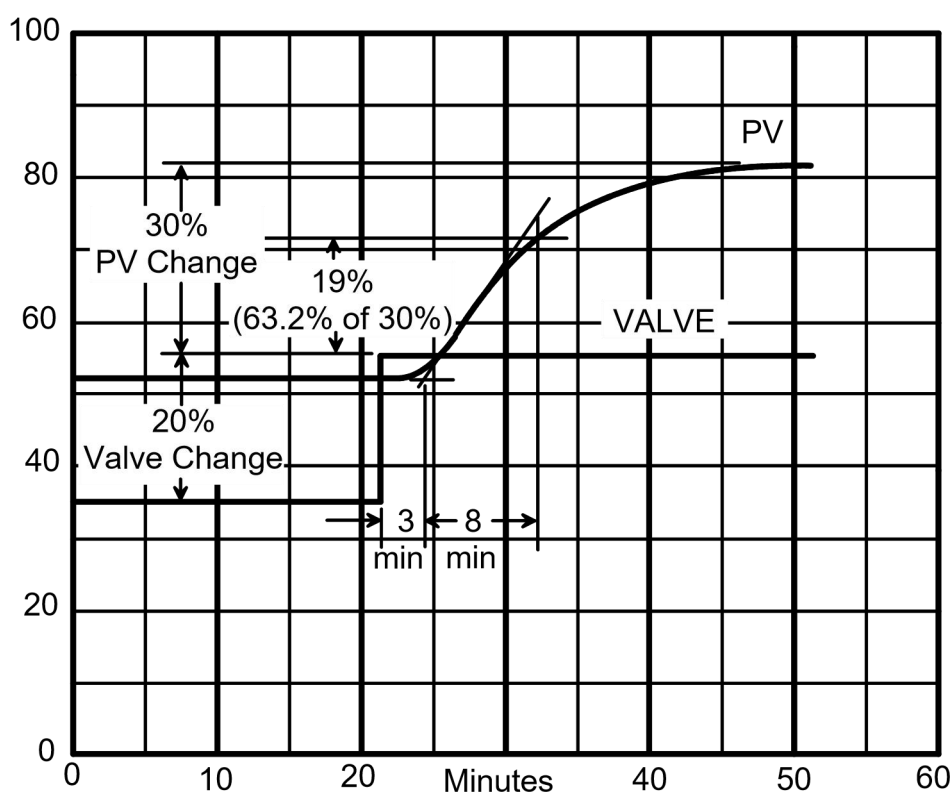


Figure 1
EXAMPLE OF ESTIMATING PROCESS MODEL PARAMETERS

4.0 OTHER FORMS OF STEP RESPONSE

This section will explore other forms of step response.

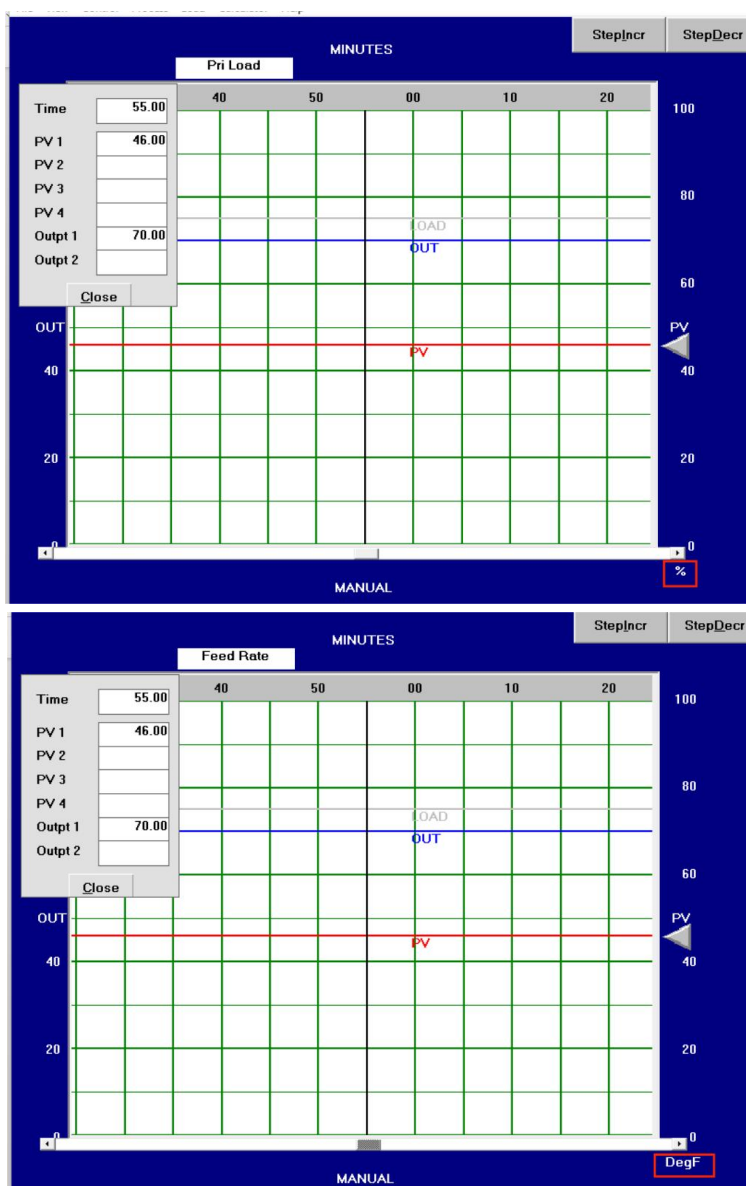
4.1 Negative Process Gain

Select **Process** | **Select Model**. Highlight "generic3.mdl" and press **Open**.

Record the initial values:

PV: 46.0 DegF 46.0 %

Controller output: 70.0 %



Increase the controller output by 10%. When the PV reaches equilibrium, record the following:

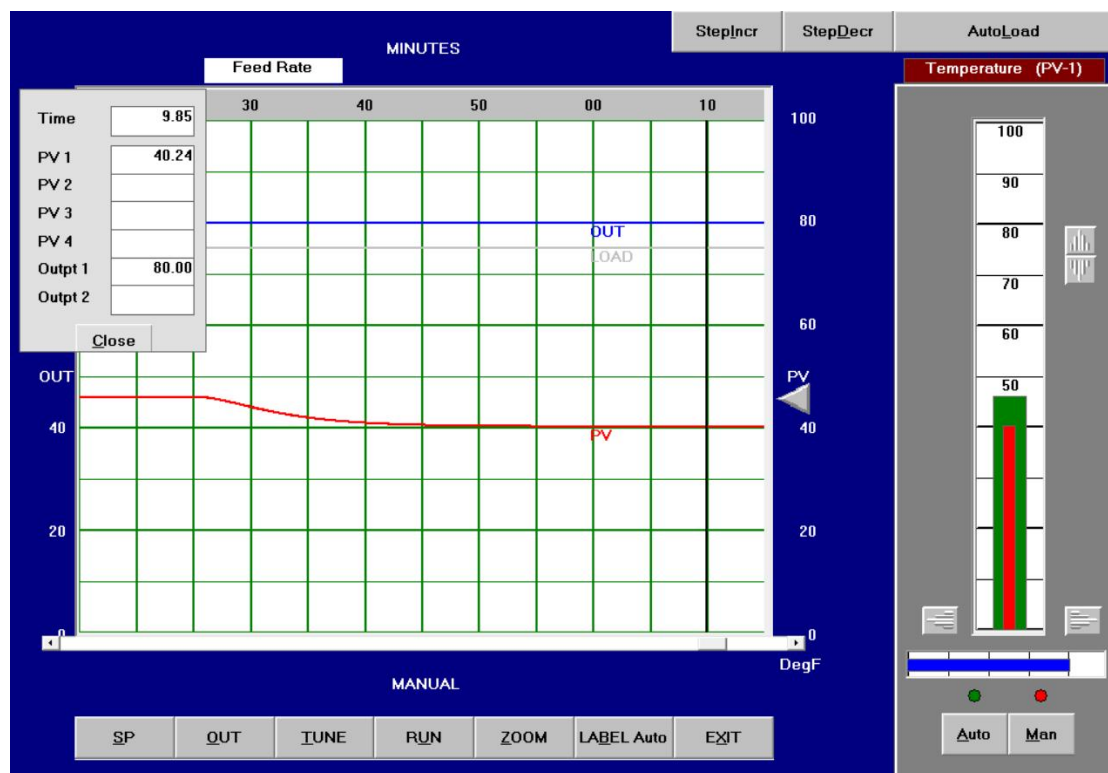
Final value of PV: 40.24 DegF 40.24 %

Final value of controller output: 80 %

Process Gain:

$$K_P = \frac{\Delta PV(\%)}{\Delta \text{Controller Output}(\%)}$$

$$K_P = \frac{(40.24 - 46)}{(80 - 70)} = -0.576 \text{ (negative process gain)}$$



4.2 Integrating Process

An **integrating process** is one which will not achieve a natural equilibrium. That is, the measurement will normally ramp up or down. The control of liquid level, where either the vessel inflow or vessel outflow is fixed by an external setpoint, is a common example of an integrating process.

Select **Process | Select Model**. Highlight "level2.mdl" and press **Open**.

This process model simulates a level control application in which the tank outflow is an independent process disturbance (load), and the controller controls a valve in the inflow line.

This process model has some elements of "realism" that are not needed in this exercise; These features will be used in subsequent laboratory exercises. For the present, we will eliminate these "realism" elements.

Select **Process | Change Parameters**.

Highlight the parameter labeled "Valve Pos: 0 = No; 1 =Yes." Change this parameter value to 1.

Select the parameter labeled "Levl Noise: 0=No; 1=Yes". (Use the scroll bar at the left-hand side of the parameter list, if this parameter is not visible initially.) Change this parameter value to 0.

Press **Close**.

BLOCK NO.	BLOCK TYPE	PARAM NO.	PARAMETER DESIGNATION	PARAM VALUE
"	"	2	0=EqPct; 1=Linear	0.0000
49	Data	1	Level Meas Range -	5.0000
50	Random	1	FlowNoise: 0=No; 1=	0.0000
"	"	2	FlowNoise Max Amp	1.0000
"	"	3	FlowNoise Correlatio	0.5
51	Data	1	Tank Diameter - Fee	3.0000
55	Gain	1	1=Cont Inflo; -1=Outfl	1.0000
57	Lead/Lag	2	Blk 57 Lead Time	1.5000
"	"	3	Blk 57 LagTime	1.5
64	Random	1	LevlNoise:0=No; 1=	0.0000

Select any line to change the parameter value

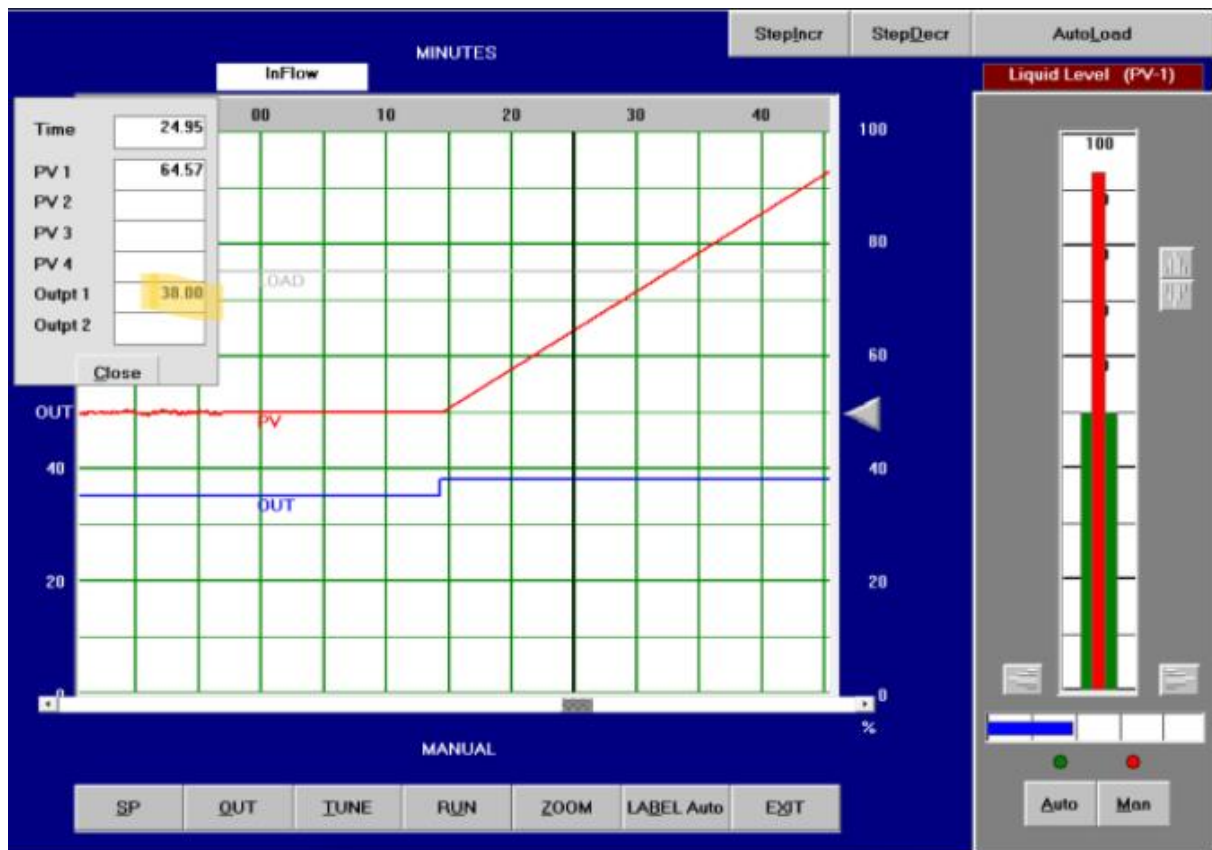
Close

BLOCK NO.	BLOCK TYPE	PARAM NO.	PARAMETER DESIGNATION	PARAM VALUE
9	Data	1	Valve Pos: 0=No; 1=	1.0000
14	Valve	1	Valve: Cvmax	51.8060
"	"	2	0=EqPct; 1=Linear	0.0000
49	Data	1	Level Meas Range -	5.0000
50	Random	1	FlowNoise: 0=No; 1=	0.0000
"	"	2	FlowNoise Max Amp	1.0000
"	"	3	FlowNoise Correlatio	0.5
51	Data	1	Tank Diameter - Fee	3.0000
55	Gain	1	1=Cont Inflo; -1=Outfl	1.0000
57	Lead/Lag	2	Blk 57 Lead Time	1.5000

Select any line to change the parameter value

Close

Change the controller output from 35.0 to 38.0. This simulates an increase in inflow to the tank.



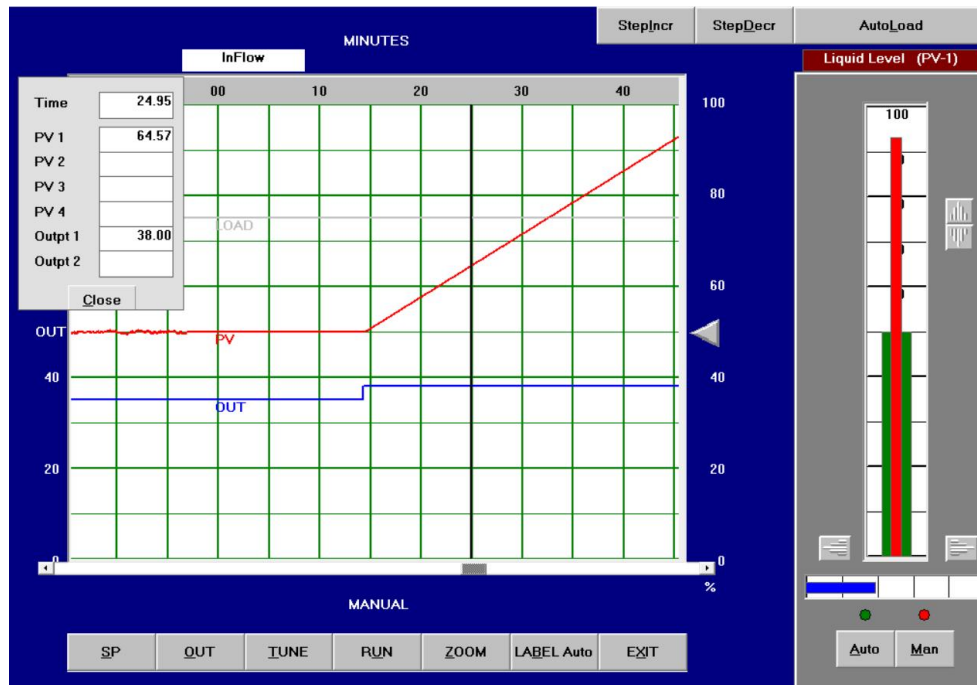
What does the PV (tank level) do? _____

The tank level increases until it reaches the limit which is 100%.

Does it appear that the PV is going to reach an equilibrium? _____

NO

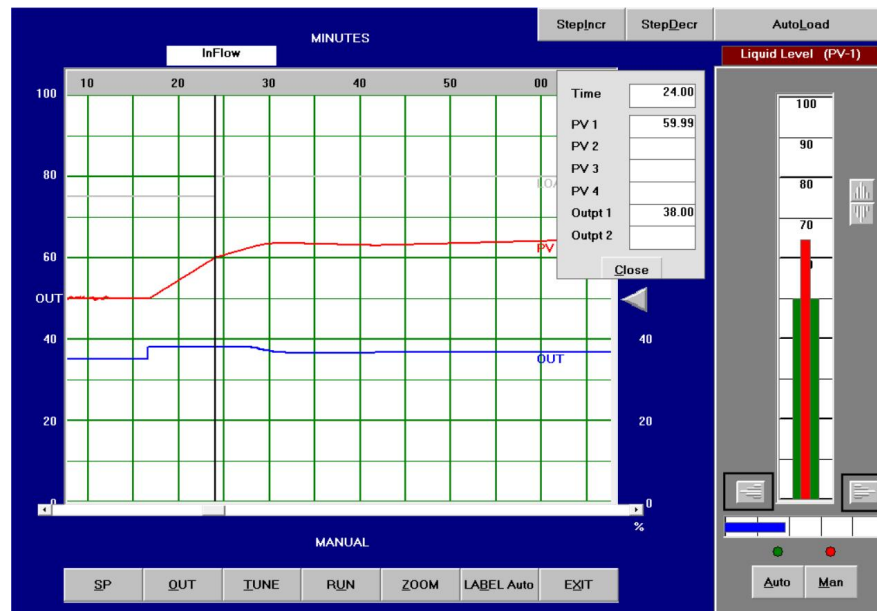
Describe the PV waveform you see The tank level increases linearly with constant rate .

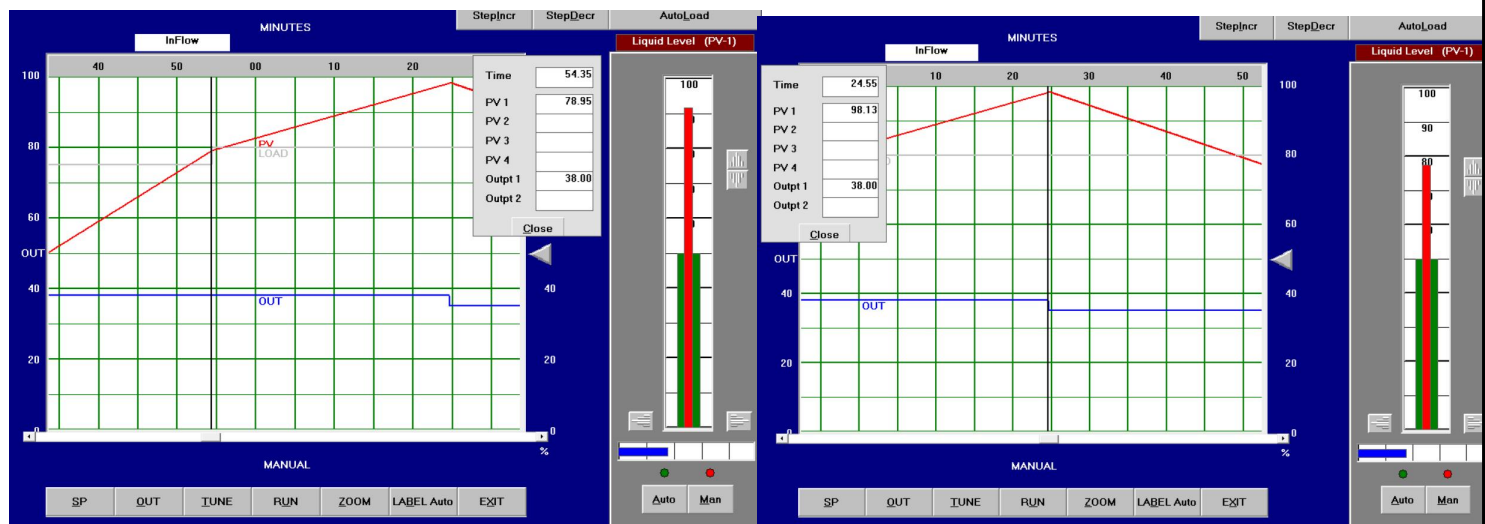


Before the PV reaches a limit (0 or 100), press the button **StepIncr** above the controller faceplate. This simulates an increase in outflow from the tank. Note the increase in the trace labeled 'LOAD'.

If the PV is now falling, then the rate of outflow must exceed the inflow. Slightly increase the controller output (to increase the tank inflow) to attempt to stabilize the level. If the PV is rising, decrease the controller output. Can you eventually (approximately) stabilize the level by manual control?

Yes, we can approximately stabilize the level by manual control, but only with continuous attention, and it will never be perfectly stable.





4.3 Runaway Process

A **runaway process** is one which is unstable without feedback control. That is, if the controller is left on manual, the measurement will accelerate as it climbs up or falls down until some physical (possibly catastrophic) limit is reached. An example is the exothermic reactor.

Select **Process | Select Model**. Highlight “runaway.mdl” and press **Open**.

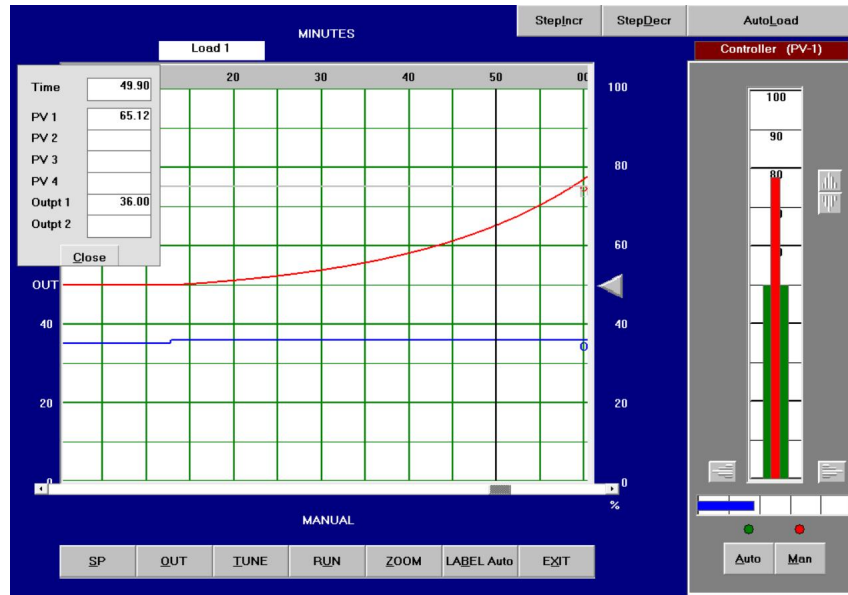
Change the controller output from 35.0 to 36.

What does the PV do? _____

The PV increased . _____

Describe the PV waveform you see

The PV increased exponentially. _____



Before the PV reaches a limit (0 or 100), try to intervene and bring it back as close as possible to 50% *by manual control*.

How easy is this task?

The task is very difficult because the response changes rapidly.

4.4 Inverse Process Response

Select **Process** | **Select Model** and load "level2.mdl" process.

Select **Process** | **Change Parameters**.

Select the parameter labeled "Levl Noise: 0=No: 1=Yes". Change this parameter value to 0.0.

Select the parameter labeled 'BLK 57 LEAD TIME". Change the value of this parameter from 1.5 to -3.0.

Press **Close**.

Press **StepIncr**. This simulates an increase in steam flow from the boiler drum.

Observe the PV (drum level). What does it initially do?

The level initially increases.

(This is the drum level "swell" effect. If the steam flow had been decreased, rather than increased, we would have seen the opposite effect, or the "shrink".)

What does the level eventually do?

The level decreases

Suppose a feedback level controller had been in Automatic and controlling the feedwater rate (drum input). Upon sensing the initial change in level, would the level controller Increase or decrease the feedwater rate?

When the steam flow increases, the drum level initially rises due to the swell effect. A feedback level controller in Automatic reacts only to the measured PV (level). Seeing the level above setpoint, it will reduce the feedwater flow to try to bring the level back down .

Would this change be in the proper direction for long-term correction?

No, this is the wrong direction for long-term correction because the level decreases .

